

GOVT. POLYTECHNIC, AMRAVATI.
(An Autonomous Institute of Govt. of Maharashtra)

TERM END EXAMINATION

WINTER - 2024

Write Identity Code

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PROGRAMME: DIPLOMA IN COMPUTER ENGINEERING

COURSE CODE & ITS TITLE: -CM5460 COMPUTER GRAPHICS

Time Allowed : 03 Hrs

Max. Marks: 70

Instructions:

1. Write your Identity Code Number on question paper.
2. All questions are compulsory.
3. Illustrate your answers with neat sketches wherever necessary.
4. Use of non-programmable calculator is permissible.
5. Figures to the right indicate full marks.
6. Assume suitable additional data, - if necessary and state the assumptions made.
7. Each sub-question in a question carries equal marks unless otherwise specified.
8. Q.1 to Q.6 are as per CO1 to CO6 respectively

Marks

Q.1. (A) Attempt any TWO.

08

- a) Write any four advantages of computer graphics in brief.
- b) Differentiate between Raster and Random scan display. (Any four points)
- c) Write any four graphics file formats. Explain any two graphics file formats.

Q.2.(A) Attempt any ONE.

02

- a) Write any two disadvantages of simple DDA line drawing algorithm.
- b) Define playgon. Enlist its types.

(B) Attempt any THREE.

12

- a) Write down scan line fill algorithm used to fill polygon.
- b) Write the steps of Bresenham's line drawing algorithm.
- ☒ c) Rasterize the line from point (1,1) to point (5,6) using DDA line drawing algorithm.
- d) Write steps of Midpoint circle drawing algorithm .

Q.3.(A) Attempt any ONE.

02

- a) Define translation.
- b) Write 3D matrix for rotation about x-axis and z-axis.

(B) Attempt any THREE.

12

- a) Define rotation. Write down homogeneous co-ordinate matrix for 2D rotation.
- b) Translate a polygon with coordinates A(2,5), B(7,10) and C(9,1) by 2 units in x-direction and 4 units in y-direction.
- c) Describe 3D translation transformation along with its matrix representation.
- ☒ d) Explain 2D transformation matrix for rotation about arbitrary point.



Q.4.(A) Attempt any ONE.

02

- a) Define viewport.
- b) Define polygon clipping.

(B) Attempt any THREE.

12

- a) Describe processing of polygon vertices in Sutherland -Hodgman polygon clipping algorithm with suitable example.
- b) Describe Cohen-Sutherland line clipping algorithm with example.
- c) Write the steps of Midpoint subdivision line clipping algorithm.
- d) Explain viewing transformation with normalization transformation and workstation transformation.

Q.5.(A) Attempt any TWO.

04

- a) Define interpolation.
- b) State any 02 properties of Koch curve.
- c) Define fractal line.

(B) Attempt any TWO.

08

- a) Write algorithm for arc generation using DDA algorithm.
- b) Explain Bezier curve with suitable diagram.
- c) Draw and explain second approximation to Hilbert's curve.

Q.6.(A) Attempt any TWO.

08

- a) Explain in brief any four hazards of graphics standard.
- ~~b)~~ With neat diagram, explain how OpenGL works.
- ~~c)~~ Describe GUI along with its any two components.

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TERM END EXAMINATION
WINTER - 2024

PROGRAMME: DIPLOMA IN COMPUTER ENGINEERING

COURSE CODE & ITS TITLE: CM5461 PROGRAMMING WITH PYTHON.

Time Allowed : 03 Hrs

Max. Marks: 70

Instructions:

1. Write your Identity Code Number on question paper.
2. All questions are compulsory.
3. Illustrate your answers with neat sketches wherever necessary.
4. Use of non-programmable calculator is permissible.
5. Figures to the right indicate full marks.
6. Assume suitable additional data, - if necessary - and state the assumptions made.
7. Each sub-question in a question carries equal marks unless otherwise specified.
8. Q.1 to Q.6 are as per CO1 to CO6 respectively

Marks

Q.1.A) Attempt any TWO.

08

- a) Explain any four features of python.
- b) List the different input output statement in python with example.
- c) Define the following terms.
 - i) Identifiers
 - ii) Indentation.

Q.2.A) Attempt any ONE.

- a) Find the output of the following code.

a=12.345

b = 3

c = 3 + 4j

print ("Type of a:", type (a))

print ("Type of b:", type (b))

print ("Type of c:", type (c))

- b) Find the output of the following code.

x = 20

y = 50

z = 20

print ("x is y:", x is y)

print ("x is z:", x is z)



Q.2.B) Attempt any TWO.

08

- a) Develop a python program to calculate the amount for the electricity bill after accepting number of units. Consumed by the user according to the following table.

Units	Price
First 200 units	No charge
Next 300 units	Rs. 5 per unit
After 500m units	Rs. 10 per unit

- b) Explain the different standard data type available in python.
c) Explain the various conditional statements along with its demonstration in python.

Q.3.A) Attempt any TWO.

04

- a) Enlist the various functions used on list in python.
b) Recall the term list comprehension with example.
c) Recall term indexing in python.

Q.3.B) Attempt any TWO.

08

- a) Explain the slicing in lists with example.
b) Develop a python program to sum of all the items in a given list.
c) Explain the following functions with dictionary.
a) dict ()
b) len ()
c) get ()
d) pop ()



Q.4.A) Attempt any ONE.

02

- a) Define the term python packages.
b) Recall the term anonymous function.

Q.4.B) Attempt any THREE.

12

- a) Describe the module in python with example.
b) Develop a python program to check if string is palindrome or not using function.
c) Develop a python program to find factorial of a given number.
d) Explain the term exceptions in python. Demonstrate the "Index Error" exceptions in python.

Q.5.A) Attempt any TWO.

04

a) Write about the following modes of files.

- i) r
- ii) r +
- iii) w +
- iv) wb +

b) List & explain various the function used for opening and closing of a file.

c) Recall the term readline () and readlines () in python.

Q.5.B) Attempt any TWO.

08

a) Develop a python to find the longest word in a file.

b) Explain the term “file” and write the various operation perform on file.

c) Demonstration the write () and writeline () functions with the help of example with respect to files in python.

Q.6.A) Attempt any ONE.

02

a) Recall the term class and object in python with example.

b) List the various features of object oriented programming.

Q.6.B) Attempt any THREE.

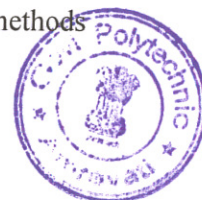
12

a) Describe the various type inheritance with help of example.

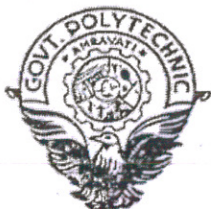
b) Develop a program to define a class named circle with an attribute for radius. Include methods for calculating the value of its area and perimeter.

c) Demonstrate how data abstraction and encapsulation are implemented in python.

d) Develop a python program to demonstrate single inheritance.



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PROGRAMME: DIPLOMA IN COMPUTER ENGINEERING

COURSE CODE & ITS TITLE : CM5465-CLOUD COMPUTING

Time Allowed : 03 Hrs

Max. Marks: 70

Instructions:

1. Write your Identity Code Number on question paper.
2. All questions are compulsory.
3. Illustrate your answers with neat sketches wherever necessary.
4. Use of non-programmable calculator is permissible.
5. Figures to the right indicate full marks.
6. Assume suitable additional data, - if necessary and state the assumptions made.
7. Each sub-question in a question carries equal marks unless otherwise specified.
8. Q.1 to Q.6 are as per CO1 to CO6 respectively



Marks

-
- | | |
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| Q.1.(A) Attempt any ONE. | 02 |
| a) State need of cloud computing. | |
| b) Define Saas service. | |
| (B) Attempt any TWO. | 08 |
| a) Describe any four features of cloud service models. | |
| b) Explain cloud deployment model. | |
| c) Compare Lass, Saas, Paas service. | |
| Q.2.(A) Attempt any TWO. | 04 |
| a) Define VM consolidation. | |
| b) Define hardware virtualization. | |
| c) List any two disadvantages of virtualization. | |
| (B) Attempt any ONE. | 04 |
| a) Describe levels of virtualization. | |
| b) Describe reference model of virtualization. | |
| Q.3.(A) Attempt any TWO. | 04 |
| a) Write any two benefits of virtual data center. | |
| b) Define file level storage virtualization in cloud. | |
| c) State need of Google file system. | |
| (B) Attempt any THREE. | 12 |
| a) Draw and explain VDC architecture. | |
| b) Explain architecture of Google file system. | |
| c) Describe advantages of GFS and HDFS. | |
| d) Describe benefits of virtual storage area network. | |

Q.4.(A) Attempt any TWO.

04

- a) List components of federated cloud.
- b) Write any two functions of cloud portal.
- c) State taxonomy of resource management.

(B) Attempt any THREE.

12

- a) Describe periodic resource optimization.
- b) Analyze the cloud service life cycle required to provide cloud services in your institute.
- c) Describe Ab-initio Resource Assignment.
- d) Describe SLA life cycle.

Q.5.(A) Attempt any ONE.

- a) State need of content level security.
- b) Define data masking.

(B) Attempt any TWO.

- a) Describe Digital identity and access management.
- b) Describe challenges of cloud security.
- c) Describe technologies used for data security.

Q.6. Attempt any ONE.

- a) Define windows AZURE cloud platforms.
- b) What is Google App Engine?

(B) Attempt any TWO.

- a) Compare the features of AWS and windows AZURE cloud platforms.
- b) Describe Amazon EC2 and its basic features.
- c) Describe steps to configure the server using Microsoft AZURE to secure it.



08

02

08

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TERM END EXAMINATION
WINTER - 2024

PROGRAMME: DIPLOMA IN COMPUTER ENGINEERING

COURSE CODE & ITS TITLE: CM5467- COMPUTER SECURITY

Time Allowed : 03 Hrs

Max. Marks: 70

Instructions:

1. Write your Identity Code Number on question paper.
2. All questions are compulsory.
3. Illustrate your answers with neat sketches wherever necessary.
4. Use of non-programmable calculator is permissible.
5. Figures to the right indicate full marks.
6. Assume suitable additional data, - if necessary and state the assumptions made.
7. Each sub-question in a question carries equal marks unless otherwise specified.
8. Q.1 to Q.6 are as per CO1 to CO6 respectively

Marks

Q.1. Attempt any TWO.

08

- a) Describe computer security with its need.
- b) List the types of attack and explain any two attacks.
- c) Differentiate between Intruders and Insiders.

Q.2.(A) Attempt any TWO.

04

- a) Define the authentication and authorization.
- b) List any four biometric methods used for identification.
- c) List any two password attack.

(B) Attempt any TWO.

08

- a) Explain criteria for password selection.
- b) List the different types of biometric mechanism (any four). Describe fingerprint mechanism for authentication user.
- c) Describe the following Access Control Policies i) DAC ii) MAC.

Q.3. Attempt any THREE.

12

- a) Explain the terms : Cryptography, Cryptanalysis and Cryptology.
- b) Explain transposition technique. Convert the below plain text into cipher text using rail fence techniques "COMPUTER SECURITY".
- c) Convert a plaintext "Cyber Security" into cipher text by using Caesar's Cipher. Consider a shift of 4 positions.
- d) Differentiate between Symmetric and Asymmetric Cryptography.